

Signalling in labour markets part 2

Marie-Louise Vierø

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Readings: MSPC 5.1-5.3

Perfect Bayesian EQ

There are two types of equilibria:

- Separating in which $e^*(\theta_L) \neq e^*(\theta_H)$.
- Pooling in which $e^*(\theta_L) = e^*(\theta_H)$.

Separating equilibria

Fact 1: In any separating equilibrium, $w^*(e^*(\theta_H)) = \theta_H$ and $w^*(e^*(\theta_L)) = \theta_L$.

Why: In a PBE, beliefs on the equilibrium path must be correctly derived from the equilibrium strategies using Bayes' rule.

Since we are in a separating equilibrium where the two types send different messages, firms must assign prob. 1 to θ_H , respectively θ_L , after seeing education level $e^*(\theta_H)$, respectively $e^*(\theta_L)$.

I.e. $\mu(e^*(\theta_H)) = 1$ and $\mu(e^*(\theta_L)) = 0$.

Separating equilibria

Fact 2: In any separating equilibrium, $e^*(\theta_L) = 0$.

Why: Since worker types separate out, low type receives wage θ_L . Thus choosing an $e > 0$ would give him more disutility but would not increase wage.

So the low type will choose $e = 0$.

Separating equilibria

Facts 1 and 2 give that we have an equilibrium where $e^*(\theta_H) = \tilde{e}$, which lies at the intersection of the indifference curve for θ_L through $(0, \theta_L)$ and the break-even line for employing type θ_H .

A low-ability worker can choose between $(0, \theta_L)$ and $(\theta_H, \tilde{e})$. $\tilde{e}$ is exactly the level of education where the low type is indifferent between the two.

Separating equilibria

The equilibrium we have derived consists of strategies

$$e^*(\theta) = \begin{cases} 0 & \text{if } \theta = \theta_L \\ \tilde{e} & \text{if } \theta = \theta_H \end{cases}$$

for the worker and

$$w^*(e) = \begin{cases} \theta_H & \text{if } e \geq \tilde{e} \\ \theta_L & \text{if } e < \tilde{e} \end{cases}$$

for the firms, and beliefs

$$\mu(e) = \begin{cases} 1 & \text{if } e \geq \tilde{e} \\ 0 & \text{if } e < \tilde{e} \end{cases}$$

Note that we have Bertrand competition between the firms, so the worker gets all of the surplus.

Non-uniqueness of equilibrium

- The equilibrium is not unique.
- Let e_1 be the education level that satisfies $w - c(e_1, \theta_H) = \theta_L$. That is, the level that makes the high type indifferent between taking the offers $(0, \theta_L)$ and (e_1, θ_H) .
- We could have picked any education level $\hat{e} \in [\tilde{e}, e_1]$ and gotten an equilibrium with

$$w^*(e) = \begin{cases} \theta_H & \text{if } e \geq \hat{e} \\ \theta_L & \text{if } e < \hat{e} \end{cases}$$

$$\mu(e) = \begin{cases} 1 & \text{if } e \geq \hat{e} \\ 0 & \text{if } e < \hat{e} \end{cases}$$

$$e^*(\theta) = \begin{cases} 0 & \text{if } \theta = \theta_L \\ \hat{e} & \text{if } \theta = \theta_H \end{cases}$$

Non-uniqueness of equilibrium

- There is a continuum of education levels for high types that can be an eqm solution. This illustrates the multiplicity of eqa.
 - Eqa can be Pareto ranked. Which eqas are Pareto dominated?
 - Why are Pareto dominated eqa sustained in the notion of PBE?
 - In eqm high ability workers are afraid to choose education levels e below the eqm level $\hat{e} > \tilde{e}$. These beliefs are never disconfirmed because in equilibrium the high ability individual choose $\hat{e}$ due to these beliefs.
 - I.e. since these information sets are never reached we can apply arbitrary eqm beliefs
 - Self-fulfilling beliefs: because of firms' strategies workers never visit out-of-eqm nodes.
- Given that nodes are never visited, beliefs are not disconfirmed.

Non-uniqueness of equilibrium

There is also multiplicity in the sense that there are different PBE's with same allocation but different out-of-equilibrium beliefs.

Pooling equilibria

Any $(e, w) = (e^*, E[\theta])$ with $e^* \in [0, e_p]$ is a pooling eqm path where e_p is the education level which makes a θ_L worker indifferent between $(0, \theta_L)$ and $(e_p, E(\theta))$ where the wage is

$$E(\theta) = \lambda\theta_H + (1 - \lambda)\theta_L.$$

An equilibrium wage schedule is

$$w(e) = \begin{cases} E(\theta) & \text{if } e \geq e^* \\ \theta_L & \text{if } e < e^* \end{cases}$$

and the corresponding eqm beliefs are $\mu(e) = \begin{cases} \lambda & \text{if } e \geq e^* \\ \theta_L & \text{if } e < e^* \end{cases}$

Pooling equilibria

- The equilibrium wage schedule is sustained by the fear of workers of delivering an unfavourable impression of ability. Therefore, in eqm beliefs are never tested and therefore they remain.
- Multiple equilibria
- Equilibria with different education levels
- Intuition for $e^* > 0$: the agent fears to be believed to be the low type if he chooses a lower education level
- Pooling eqm with no education Pareto dominates the other pooling eqa.
- But note also that the same education level can be supported by different belief functions.

Summary

- Separating eqm paths:
 - Low type: $(e, w) = (0, \theta_L)$.
 - High type: $(e, w) = (e^*, \theta_H)$ with $e^* \in [\tilde{e}, e_1]$.
 - $\tilde{e}$ determined from binding IC constraint for low type.
 - e_1 determined from binding IC constraint for high type.
- Pooling eqm paths:
 - $(e, w) = (e^*, E[\theta])$ with $e^* \in [0, e_p]$.
 - e_p determined from binding IC constraint for low type (point which makes low type indifferent between separating eq and pooling eq)
- Appropriate beliefs and wage functions are also part of the equilibrium.

Welfare compared to a world without signalling device

- Compared to a situation where no signalling mechanism is available
- Type θ_L workers are worse off because of the existence of a signalling device.
- Type θ_H workers may be worse off or better off.
- So signalling can be bad for both types of workers.

Efficiency

- Since education is “unproductive” (does not directly improve profits), the first best is more efficient than a separating equilibrium with signalling.
- I.e. as before we generally see that the first best is more efficient than the second best.
- Again the concept of “constrained pareto optimal” allocations is relevant: i.e. would a social planner prefer other allocations than what the market achieves if the planner also cannot see the type of agent he is dealing with.
- The separating eqm with $\tilde{e}$ is potentially constrained Pareto optimal (CPO), all the other separating PBEs are not.

Welfare in the second best solution

- Whether $\tilde{e}$ is CPO depends on whether a pooling eqm is Pareto improving.
- Low ability worker prefers a pooling eqm (where $e < e_P$) to the signalling eqm.
- Whether high ability workers are better off with signalling depends on λ .
- Why?

A central authority who cannot observe worker types may be able to achieve a Pareto improvement relative to the market outcome

- λ affects $E[\theta]$ and thus whether θ_H prefers the pooling equilibrium to the separating equilibrium.
- Suppose λ is “high enough”. In this case there is room for policymakers to change the equilibrium from a separating to a pooling by banning education.
- In fact this may also be possible when the the pooling outcome does not Pareto dominate the separating outcome
 - If the government by law demands contracts $(0, \hat{w}_L)$ and $(\hat{e}_H, \hat{w}_H)$ AND the contracts imply that firms break even on average
 - The key here is cross-subsidization

Equilibrium refinements

In general, many equilibria might exist in a signalling game

- Need refinements
- Problem is that beliefs off-the-equilibrium path are “arbitrary” (we saw examples of that with different wage functions above)
- Idea behind most refinements: Can we somehow restrict out-of-equilibrium beliefs? Can we require “rational” beliefs off the equilibrium path?

We look at two refinements:

- Dominance (eliminates dominated actions)
- Intuitive criterion (eliminates actions dominated in equilibrium)

Dominance criterion

Definition:

- In a signalling game message m_i is dominated for type θ_i if there exists another message m'_i such that θ_i 's lowest possible payoff from m'_i is greater than θ_i 's highest possible payoff from m_i :

$$\min_a U_S(\theta_i, m'_i, a) > \max_a U_S(\theta_i, m_i, a)$$

- **Dominance criterion (weak version):** If m_i is dominated for type θ_i then (if possible) the Receiver's belief $\mu(\theta_i|m_j)$ should place zero probability on θ_i .
- No matter what the receiver does, a sender of type θ_i would never benefit from sending m_i .

Dominance criterion

- **Dominance criterion (strong version):** If m_j is dominated for type θ_i for any possible equilibrium responses by the receiver, then (if possible) the receiver's belief $\mu(\theta_i|m_j)$ should place zero probability on θ_i .
- This does get rid of some of the PBE, since for any belief, the equilibrium wage must lie between θ_L and θ_H .
- Consider f.x. equilibrium with $e^*(\theta_H) = e_1$. Firms need to believe that any worker with education below e_1 has a positive probability of being θ_L .
- But education levels above $\tilde{e}$ are dominated for a type θ_L worker by the education level 0.
- Thus a "reasonable belief" has $\mu(\theta_H|e) = 1$ for all $e > \tilde{e}$.
- Then no separating equilibrium with $e^*(\theta_H) > \tilde{e}$ can survive.

Intuitive criterion

Definition:

- Message m_i is equilibrium dominated for type θ_i , if

$$U^*(\theta_i) > \max_a U_S(\theta_i, m_i, a),$$

where $U^*(\theta_i)$ is θ_i 's equilibrium payoff.

- **Intuitive criterion:** The receiver's belief $\mu(\theta_i|m_i)$ should place zero probability on θ_i after observing message m_i if m_i is equilibrium dominated for type θ_i .
- The set of possible actions does not include strictly dominated actions (actions sender would never choose regardless of his beliefs).

Comments:

- If we can find players who are never better off than the equilibrium payoff by deviating we should not expect them to deviate
- Note that we place no restrictions on beliefs for types that are not equilibrium dominated (which will matter in cases with more than two types).
- This refinement gets rid of all pooling equilibria and most of the separating equilibria. (Only one left is the one with education level $\tilde{e}$ for the high type.)